Gating AI Deployment on Environmental Evidence Quality

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Abstract

Carbon-aware governance proposals gate AI deployment on impact magnitude: a carbon budget, a readiness level, or a risk rating. None of them asks whether the carbon number itself can be trusted. We present EVIDENCEGATE, an environmental governance module deployed inside an open-source AI governance platform, that makes measurement provenance a first-class release criterion: a provenance-derived confidence score crossed with an impact tier resolves to *go*, *conditional_go*, or *no_go*, and blocks the platform's deployment-approval workflow. In a measurement study over four workloads, five iterations each, with every physical run measured by both a software power tracker and an analytic FLOP model, the release decision changed on **20 of 20 paired runs** while the impact tier stayed *low* throughout: the same run was released unconditionally under one measurement method and held for mitigation under the other, purely because of how the number was obtained. The two methods disagreed on magnitude by 123 percent MAPE. A mitigation experiment sharpened the point in the opposite direction: quantising a fine-tune to `bfloat16` on a host with no optimised kernel made it 46 times slower and burned roughly 40 times the energy, so a routine "we quantised, therefore we cut carbon" claim was false by an order of magnitude in the harmful direction. Evidence quality, not only impact magnitude, decides whether an environmental claim can carry a release.

1 The gap: a measurement layer with no trust layer

Regulators and standards bodies are building the measurement layer for AI's environmental footprint: quantification-methods inventories [1], sustainability overviews [2], and a G7-commissioned inventory cataloguing thirteen measurement, monitoring, and reporting initiatives [3]. The EU AI Act anticipates harmonised standards for "reporting and documentation processes to improve AI systems' resource performance" (Article 40), and general-purpose AI documentation must state known or estimated energy consumption (Annex XI) [4]. Yet none of these instruments defines how the *quality* of an environmental claim should condition a deployment decision.

Recent governance research gates on magnitude. Carbon-aware governance gates embed carbon budgets in governance layers, recording provenance only in audit logs [5]. Sustainability-pyramid frameworks gate on readiness levels [6], and risk-management integrations treat AI carbon as a risk dimension in banking frameworks [7]. Meanwhile measurement research has established that reliability varies sharply with disclosure: metadata-completeness tiers across 5,227 public models span roughly $1.2\times$ to $2\times$ estimation error [8], and spend-based fallbacks in tiered Scope 3 accounting

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overestimate by 10–40$\times$ [9]. Adjacent work has begun to act on estimate trust, through conformal calibration inside a carbon budget gate [10] and confidence-gated filtering of industrial emissions predictions [11]. The two literatures have not been connected: a system can sit comfortably under its carbon budget while its carbon number is untrustworthy, and every published gate would pass it.

Our claim is narrow and, to our knowledge, unheld: environmental *evidence quality*, not only impact magnitude, should gate AI deployment. The intellectual ancestry is the data-quality tradition in life-cycle assessment, where pedigree matrices and data quality indicators grade the provenance of an inventory figure [12], and the NUSAP uncertainty pedigree [13]. Recent work recommends indicator-style disclosure for AI-assisted product footprinting but proposes no score, no thresholds, and no gate [14]. We invert that tradition: where it assesses AI that produces carbon numbers, we score carbon numbers to gate the AI.

2 Mechanism

Provenance classes and confidence. Every assessment declares its measurement source, normalized to one of five provenance classes with bounded confidence ranges: *measured* (hardware telemetry; 0.85–1.00), *tool_estimated* (software power trackers; 0.60–0.85), *vendor_reported* (cloud-provider figures; 0.40–0.60), *manual* (analytic FLOP estimates; 0.20–0.40), and *unknown* (0.0). A caller-supplied score is honoured but never allowed to exceed its provenance cap, and unknown provenance forces 0.0 regardless of any claimed score. Provenance is categorical and reported separately from quantified uncertainty, so a precise-looking number cannot launder a weak source.

Tiers and the decision matrix. Metrics map to low, medium, or high impact per lifecycle phase, using location-based carbon so market instruments cannot lower a gate outcome. Confidence bands (measured ≥ 0.70; estimated 0.35–0.69; unknown < 0.35) cross with impact tiers into the matrix shown in Figure 1(b). Confidence 0.0 short-circuits to *no_go* before matrix lookup: an undisclosed claim never passes. A *conditional_go* cannot be approved until it carries a dated, owned mitigation or an explicit exception with owner, expiry, and rationale, which makes reviewer overrides auditable rather than informal. Six controls (ENV-1 to ENV-6: boundary, energy accounting, dual location and market carbon, per-functional-unit efficiency, evidence confidence with disclosed uncertainty, and mitigation or exception) define coverage. Disclosed offsets are recorded but never subtracted from emissions and never improve any gate output.

3 Implementation

EVIDENCEGATE ships inside an open-source AI governance platform, not as a standalone proposal. The decision engine is a framework-agnostic Python package with 57 unit tests covering every matrix cell, band boundary, provenance cap, and the undisclosed-evidence short-circuit. Assessments persist as versioned governance evidence, and a release gate hooks the platform's existing deployment-approval workflow, returning HTTP 409 with the blocking reason when the latest assessment is *no_go* or an unmitigated *conditional_go*. Reviewers get an interface showing the matrix position, control coverage, mitigation tracker, and version trail, and disclosures export in a format shaped to the European sustainability reporting standard for climate change [15].

On a 14-case hand-labelled fixture set spanning all matrix cells, mitigation states, exception paths, and offset disclosures, the full mechanism reproduces 14/14 expected labels, while ablations degrade sharply: a carbon-only gate manages 7/14, a generic sustainability score 6/14, and no environmental gate 3/14, with removing only the exception path or the mitigation review costing one to three cases each. The authors wrote both the fixtures and the engine, so this is a consistency illustration showing the confidence dimension is load-bearing, not an empirical validation. That is what the next section is for.

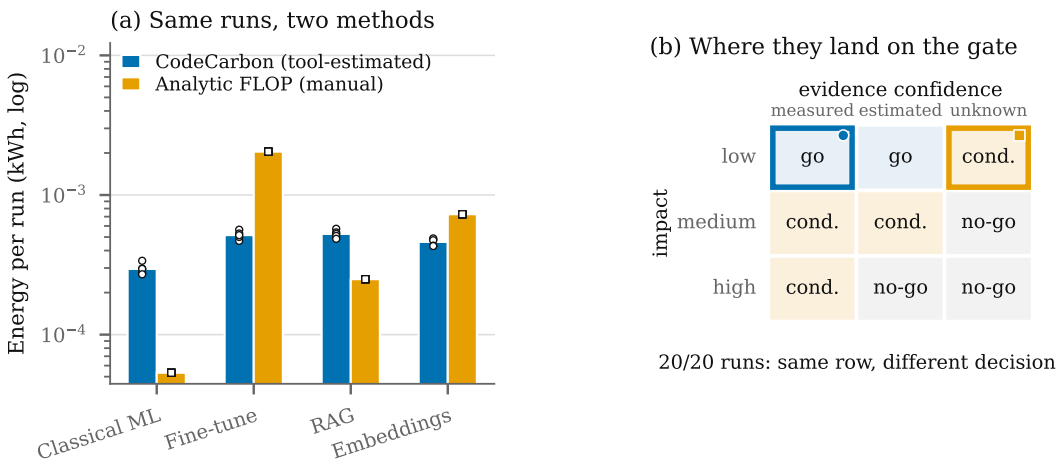


Figure 1: Four workloads, five iterations each, every physical run measured twice. (a) The two methods disagree on energy by 123 percent MAPE, and not in a consistent direction. (b) The production decision matrix; outlined cells mark where the study's runs landed. Both methods put every run in the same *low* impact row, so magnitude drove nothing: the column moved because provenance moved, and with it the release decision.

4 Measurement study: provenance decided the release

Four workloads (classical ML training, transformer fine-tune, RAG inference, batch embeddings) ran five iterations each on a single host. Every physical run was measured three ways: a software power tracker (*tool_estimated*), operator-filled cloud billing (*vendor_reported*), and a documented analytic FLOP model (*manual*). The harness never fabricates: cloud billing lags execution by hours to days, so its 20 records are explicit unavailable entries with a stated reason rather than imputed numbers. All runs were replayed through the production engine, not a reimplementation.

The decision changed on 20 of 20 paired runs. The power tracker mapped to *tool_estimated*, confidence 0.72, the *measured* band, and *go*; the FLOP model mapped to *manual*, confidence 0.30, the *unknown* band, and *conditional_go*. The impact tier was *low* in every case (Figure 1(b)), so nothing about the magnitude produced the difference. The same physical run was released unconditionally under one measurement method and held for mitigation under the other, purely because of how the number was obtained. A carbon-budget gate or a readiness-level gate sees *low* impact in both columns and passes both.

Method agreement is poor in its own right: MAPE 122.6 percent with the tracker as reference and 169.1 percent with the FLOP model as reference, ranging from 52 to 298 percent by workload, with Bland-Altman limits of agreement spanning -1.15×10^{-3} to 1.79×10^{-3} kWh.

Mitigations, and a claim that was false in the harmful direction. Four interventions were measured as pre/post pairs (Figure 2). The pre-registered test that mitigations reduce intensity by more than 20 percent is **not supported**: paired $t = -0.859$, $p = 0.410$ over 11 pairs, mean reduction -311 percent. Ten of eleven pairs did reduce, with a median of 33.9 percent. One pair inverted the mean: quantising the fine-tune from `float32` to `bfloat16` made it 46 times slower, because the host had no optimised reduced-precision kernel, and burned roughly 40 times the energy. Reporting only the median, or dropping the outlier, would be cherry-picking, so the pre-registered result stands as computed.

This is the mitigation analogue of the paper's argument. "We quantised the model, therefore we cut carbon" is a routine and entirely plausible claim; on this hardware it was false by more than an order of magnitude, in the harmful direction. A gate that accepts asserted mitigations would have recorded

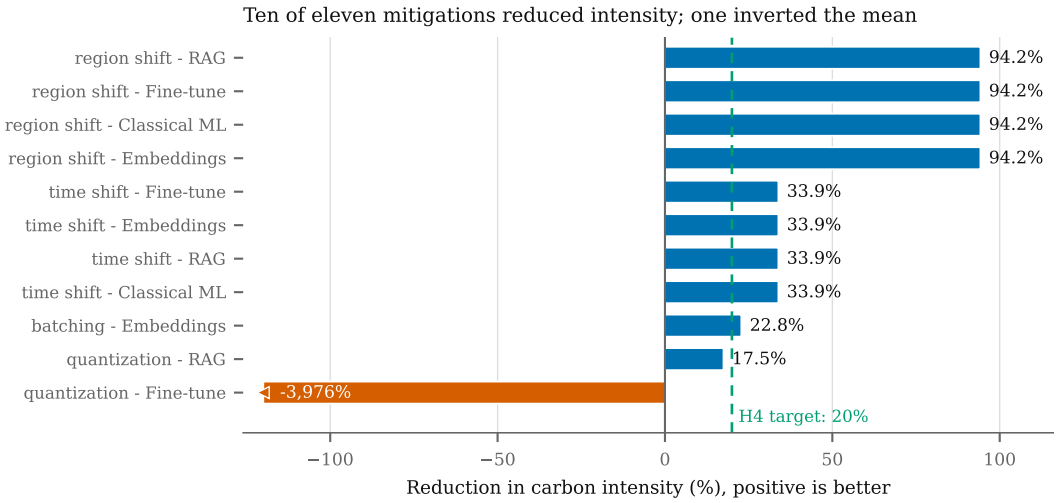


Figure 2: Mitigation outcomes. Ten of eleven pairs reduced carbon intensity, four of them past the 20 percent target. One inverted the mean: quantising the fine-tune from `float32` to `bfloat16` on a host with no optimised kernel took it from 62.7 s to 2887.6 s and burned roughly 40 times the energy. That bar is truncated and labelled with its true value.

a reduction that never happened, which is why ENV-6 requires the mitigation to be documented and re-measured rather than declared.

Expert panel (registered, in progress). Twelve engine-validated profiles covering all nine matrix cells are rated by 15 reviewers across governance, ML engineering, and sustainability roles under a two-arm counterbalanced design (baseline: raw measurements only; module: full EVIDENCEGATE output), testing whether displayed confidence correlates with conservatism, whether the module arm raises ENV control coverage, and whether the matrix improves inter-rater agreement. Instrument and analysis are implemented; we seek critique of the baseline arm before collection closes.

5 Pathway to climate impact

Environmental reporting for AI is moving from voluntary to mandatory, but mandatory reporting only reduces emissions if the reported numbers are trustworthy enough to act on. Our measurement study shows the gap is not hypothetical: two standard methods applied to the same run disagreed by more than 100 percent and implied opposite release decisions, and a textbook efficiency intervention increased emissions fortyfold while a declaration-based gate would have logged it as a reduction. Making evidence quality a release criterion changes the incentive directly: a team that cannot substantiate its number cannot ship unconditionally, so measurement effort becomes the cheapest path to approval rather than an unrewarded cost. This is a leverage point rather than direct abatement: it does not itself cut emissions, it makes the reductions others report verifiable and therefore actionable. The near-term channel is regulatory: Article 40's harmonised standards for resource-performance reporting do not yet exist [4, 16], and a provenance-scored, platform-integrated gate is a concrete, testable proposal for what they could require.

Limitations. The study host exposes no hardware energy counters, so the tracker fell back to load-based estimation: this study compares an estimate against an estimate and never claims the *measured* provenance class. These are benchmark workloads, so only the low-impact row is exercised and the provisional thresholds cannot be recalibrated from this data. The FLOP model uses placeholder efficiency constants, and the panel targets 15 raters, sized for feasibility rather than power. Code, data, and figure scripts are public.

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